

Distribution of micro- and macro-elements in anaerobic co-digestion of sewage sludge and food processing wastes

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Abstract: Recently, anaerobic co-digestion has become an indispensable strategy for improving the effectiveness of municipal sewage sludge (MSS) mono-digestion. This technology is recognised as a disposal method for organic wastes with renewable energy generation and production of nutrient-rich digestate. The food processing wastes are among the most recommended substrates. Its introduction to MSS might provide valuable micro- and macro- elements which can result in increased methane production and improved quality of digestate. In this study, micro- and macro-elements distribution in anaerobic co-digestion of MSS and food processing wastes represented by ice-cream processing residues (IC), apple pomace (AP) and orange wastes (OW) has been evaluated. The batch anaerobic digestion experiments at temperature range of 37°C were conducted differing feedstock composition. In the feedstock and digestate the following metals were analysed: Na, Mg, P, K, Ca, S, Fe, Cr, Mn, Ni, Cu, Zn, As, Cd, Hg, Pb. In almost all cases, the introduction of co-substrates to MSS resulted in the improvement of macro-elements contents, simultaneously the reduction of the heavy metals (HMs) content was observed. The most favourable results were observed in three-component mixture in the presence of MSS, IC and OW, therein enhanced methane production of 401.6 Ncm³·g⁻¹ VS (volatile solids) accompanied by improved organic compounds removal was noticed. The opposite trend, i.e. decreased methane production, was obtained in the reactor supplied by AP, therein also the highest bioaccumulation of almost all HMs occurred indicating the deterioration on process efficiency. The composition of all digestates meets the requirements for MSS intended for agricultural use.

Keywords: anaerobic digestion, ecological risk, food wastes, heavy metals, sewage sludge

INTRODUCTION

In recent years, anaerobic co-digestion has become an essential strategy for enhancing the effectiveness of municipal sewage sludge (MSS) mono-digestion. This technology is recognised as a desirable disposal method for various organic wastes owing to renewable energy generation e.g. biogas, production of valuable products e.g. nutrient-rich digestate. The most preferred group of substrates are food processing wastes represented by fruit and vegetable pulps, spent grains from breweries and distilleries, by-products from dairy processing as well as oilseed cakes (Karki *et al.*, 2021; Ferdeş

et al., 2023). Their introduction to MSS might significantly improve feedstock composition through supplementation of nutrients and various micro-elements, resulting in enhanced methane production (Ferdeş *et al.*, 2023). Another important factor supporting the use of this strategy is dilution of toxic compounds, including heavy metals (HMs). This fact might also positively influence process performance. Moreover, it might provide higher quality digestate for agricultural applications (Dębowski *et al.*, 2022; González, Peña and Gómez, 2022). Importantly, the digestate obtained from MSS mono-digestion might lead to several environmental issues including enhanced soil

salinity, eco- and phyto-toxicity as well as HMs bioaccumulation (Kupper *et al.*, 2014). In this context, the variation of HMs within anaerobic bioconversion should be further investigated (Zhang *et al.*, 2022; Pilarska *et al.*, 2024). Generally, the impact of HMs on anaerobic digestion (AD) differs significantly, their influence can be both stimulating and inhibiting, depending on the form and concentrations. Moreover, from the standpoint of further agriculture utilisation of MSS, its determination in digestate might play a crucial role (Zhang *et al.*, 2022; Pilarska *et al.*, 2024).

In this study, micro- and macro-elements variation in anaerobic co-digestion of MSS and food processing wastes, represented by ice-cream processing residues (IC), apple pomace (AP) and orange wastes (OW) was evaluated. The effectiveness of process performance including methane production has also been examined. The work also identifies the ecological risk potentially associated with the use of digestate. Thus far, such evaluation has not been previously subject of the scientific research. Importantly, the obtained results might allow for effective management of food industry wastes in municipal wastewater treatment plants with energy and nutrient recovery.

MATERIAL AND METHODS

PREPARATION AND CHARACTERISATION OF THE APPLIED SUBSTRATES

The main component was represented by municipal sewage sludge (MSS) that was a mixture of primary and excess sludge. Both were separately obtained from municipal wastewater treatment plant (MWWTP), which involves the mechanical and biological steps. The samples were mixed under laboratory conditions at the volumetric ratio of 40:60 using low speed homogeniser.

Orange wastes (OW) consisted mainly of orange peels obtained as solid waste from juice production (Photo 1a). Before supplying the digesters, it was ground using laboratory blender to the size of 5 mm.

Ice-cream processing residues (IC) and apple pomace (AP) were obtained from companies operating in the food industry. The first one (Photo 1b) is a waste product from the ice-cream production process. The second one (Photo 1c) is a waste generated during the processing of fruit for jam production. These substrates were not subjected to any additional processing before being put into co-fermentation.

The inoculum for batch experiments were obtained from the same MWWTP. This sample was obtained as the effluent

from mesophilic digesters. After collection, it was incubated anaerobically at $37 \pm 0.2^\circ\text{C}$ for at least 14 days to achieve non-specific biogas production. It was characterised by total solids (TS) of $24.5 \pm 5.8 \text{ g}\cdot\text{kg}^{-1}$, volatile solids (VS) of $14.4 \pm 3.7 \text{ g}\cdot\text{kg}^{-1}$ and pH of 7.36 ± 2.1 . The detailed composition of substrates are shown in Table 1.

EXPERIMENTAL DESIGN

In this study, four experimental trials were provided. In the first one (S1), the control, the mono-digestion of MSS was performed. In trials S2 and S3, MSS was supplemented by IC and AP, respectively. In S4, three-component co-digestion of IC, OW and MSS was conducted. The two-component co-digestion of OW and MSS was the subject of the author's previous research (Szaja, Montusiewicz and Lebiocka, 2023). The adopted doses and operational set up are shown in Figure 1.

The anaerobic experiments were performed in batch mode using a laboratory device provided by BPC Instruments (Sweden). It consist of six digesters with the active volume of 2 dm^3 equipped with an agitation system placed in a water bath to maintain the adopted temperature conditions. The biogas production was continuously monitored using low gas volume and a flow meter provided by the same company. Each digester was supplied with 1.4 dm^3 of inoculum and 0.4 dm^3 of MSS. In co-digestion trials, MSS was supplemented with the adopted dose of selected food processing waste.

The adopted doses of applied wastes were established based on the substrate-to-inoculum ratio (S:I) to obtain the highest possible methane production. In this study, this S:I ratio varied between 0.68 and 0.74. In turn, the total organic load varied from 13.61 to $14.85 \text{ g VS}\cdot\text{dm}^{-3}$ depending on the feedstock composition. Each experimental trial lasted 21 days and was repeated three times. The average values were presented below.

ANALYTICAL METHODS

In the feedstock and digestate, the following metals were analysed: Na, Mg, P, K, Ca, S, Fe, Cr, Mn, Ni, Cu, Zn, As, Cd, Hg, Pb. An Agilent 8900 ICP-MS Triple Quad inductively coupled plasma mass spectrometer (Agilent, Santa Clara, CA, USA) was used for their determination. Before the analysis, the acid digestion of the obtained samples was conducted using a Topex Preekem microwave mineraliser. The He mode with the flow rate of $5.5 \text{ cm}^3\cdot\text{min}^{-1}$ was applied for determination of the majority

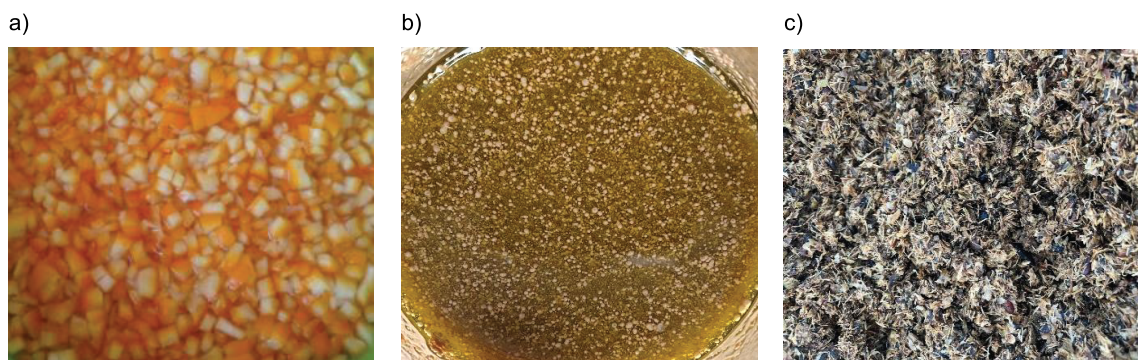


Photo 1. Substrates used for the co-fermentation process: a) orange wastes (OW), b) ice-cream processing residues (IC), c) apple pomace (AP) (phot. J. Czarnota)

Table 1. The characteristic of the co-substrates applied in the experiment

Parameter	Unit	OW	IC	AP
		mean $\pm$ SD		
TS	g·kg ⁻¹	237.60 $\pm$ 8.9	989.5 $\pm$ 11.7	105.5 $\pm$ 9.8
VS	g·kg ⁻¹	225.10 $\pm$ 10.1	979.6 $\pm$ 10.6	104.2 $\pm$ 8.7
Na	g·kg ⁻¹ DM	0.27 $\pm$ 0.01	0.35 $\pm$ 0.04	0.31 $\pm$ 0.01
Mg	g·kg ⁻¹ DM	0.77 $\pm$ 0.01	0.88 $\pm$ 0.05	0.91 $\pm$ 0.05
P	g·kg ⁻¹ DM	0.88 $\pm$ 0.02	1.12 $\pm$ 0.07	1.23 $\pm$ 0.03
K	g·kg ⁻¹ DM	10.10 $\pm$ 0.4	12.1 $\pm$ 1.3	14.1 $\pm$ 1.2
Ca	g·kg ⁻¹ DM	12.30 $\pm$ 0.3	15.6 $\pm$ 2.1	14.3 $\pm$ 2.1
S	g·kg ⁻¹ DM	0.57 $\pm$ 0.01	0.95 $\pm$ 0.07	1.3 $\pm$ 0.01
Fe	mg·kg ⁻¹ DM	180.00 $\pm$ 8.7	147.1 $\pm$ 12.4	187.2 $\pm$ 7.7
Cr	mg·kg ⁻¹ DM	0.36 $\pm$ 0.02	0.15 $\pm$ 0.04	0.25 $\pm$ 0.01
Mn	mg·kg ⁻¹ DM	6.47 $\pm$ 0.7	4.45 $\pm$ 0.07	5.54 $\pm$ 0.5
Ni	mg·kg ⁻¹ DM	0.78 $\pm$ 0.03	0.04 $\pm$ 0.001	0.07 $\pm$ 0.001
Cu	mg·kg ⁻¹ DM	12.10 $\pm$ 1.1	9.4 $\pm$ 1.2	8.7 $\pm$ 1.2
Zn	mg·kg ⁻¹ DM	29.30 $\pm$ 2.9	11.5 $\pm$ 0.9	14.3 $\pm$ 1.2
As	mg·kg ⁻¹ DM	0.014 $\pm$ 0.0001	0.004 $\pm$ 0.0001	0.008 $\pm$ 0.0001
Cd	mg·kg ⁻¹ DM	0.006 $\pm$ 0.0001	0.006 $\pm$ 0.0001	0.007 $\pm$ 0.0001
Hg	mg·kg ⁻¹ DM	0.003 $\pm$ 0.0001	0.002 $\pm$ 0.0001	0.004 $\pm$ 0.0001
Pb	mg·kg ⁻¹ DM	0.01 $\pm$ 0.0001	0.003 $\pm$ 0.0001	0.002 $\pm$ 0.0001

Explanations: OW = orange wastes, IC = ice-cream processing residues, AP = apple pomace, SD = standard deviation, TS = total solids, VS = volatile solids, DM = dry mater.

Source: own elaboration.

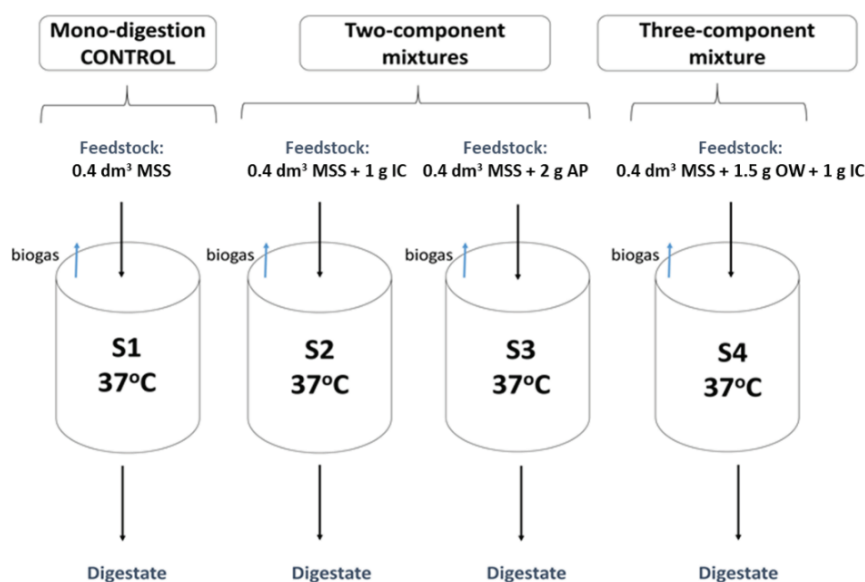


Fig. 1. The operational set-up in experiments; MSS = municipal sewage sludge, IC = ice-cream processing residues, AP = apple pomace, OW = orange wastes; source: own elaboration

metals. The O₂ mode with flow of 30 cm³·min⁻¹ was used for the As and S examination.

To evaluate the bioaccumulation of micro- and macro-elements in anaerobic digestion (AD) process the B_{index} were established according to the Equation (1):

$$B_{\text{index}} = \frac{C_D - C_F}{C_F} \cdot 100\% \quad (1)$$

where: B_{index} = biodegradability index, C_F = content of selected macro- or micro-element in the feedstock (mg·kg⁻¹ DM);

C_D = content of selected macro- or micro-element in the digestate ($\text{mg}\cdot\text{kg}^{-1}$ DM).

To evaluate the anaerobic co-digestion efficiency, the removal of organic compounds with regard to volatile solids (VS), as well as biogas and methane productions were investigated. The TS and VS contents were established based on the procedure presented in the Standard methods for the examination of water and wastewater (APHA, 2012). The stability parameters i.e. volatile fatty acid (VFA), ammonium nitrogen (AN), alkalinity (ALK) was controlled involving spectrophotometer Hach Lange DR 3900 in the supernatant sample. In turn pH-value was measured using pH meter Elmetron CP-411. The methane content was determined using a ThermoTrace GC-Ultra gas chromatograph (Thermo Fisher Scientific, Milan, Italy), equipped with a conductivity detector fitted with divinylbenzene (DVB)-packed columns.

ANOVA was applied to conduct the statistical analysis involving the Statistica software, the significance level was established at $p < 0.05$.

RESULTS

DISTRIBUTION OF MICRO- AND MACRO-ELEMENTS WITHIN THE AD PROCESS

Table 2 shows the contents of macro-elements within anaerobic digestion process. As compared to the control, the introduction of co-substrates resulted in increase of all analysed macro-elements content in the feedstock, which was related with introduction of co-substrates rich in many valuable macro-elements (Tab. 1).

Nevertheless, the significant growth varied between 36–70% was found in three-component mixtures (S4). Therein, as compared to control, the highest improvement was found with regard to Na and Mg.

Between two component trials (S2 and S3) there were no significant differences in most of analysed macro-nutrients. The exception was found in the case of S and Mg in S3, there in their contents were improved by 28 and 15% as compared to S2, respectively.

Regarding digestate composition the increased content of all analysed macro-elements occurred. As compared to control (S1) the major growths were found for S3 and S4, respectively (Tab. 2). For those trials the significant improvements exceeding 40% were found in almost all analysed cases. The exception was observed only in the case of K, where the diminished content was found in S2 and S4, respectively. As it is shown in Figure 2a, increment of all analysed macro-nutrients might be found within anaerobic digestion (AD). The major bioaccumulation was found with regard to Na and K, therein the B_{index} varied between 115–275%. In turn, as compared to control (S1) the major improvement in macro-elements contents was noticed in S3, where co-digestion of municipal sewage sludge (MSS) and apple pomace (AP) was performed. The enhancements varied between 34–64% as compared to S1.

With regard to heavy metals (HMs), the introduction of co-substrates led to their statistically significant reduction in feedstocks (Tab. 3). The exception was noticed for Hg, therein there were no significant differences between mono- and co-digestion. As compared to the control (S1), the greatest decrease in the HMs content was obtained in S4, therein the three-component co-digestion was conducted. In this case, significant

Table 2. The distribution of macro-elements in experimental trials S1–S4 ($\text{g}\cdot\text{kg}^{-1}$ DM)

Macro-element	Description	S1	S2	S3	S4
		mean $\pm$ SD			
Na	F	9.20 $\pm$ 1.3a	10.08 $\pm$ 1.2a	10.50 $\pm$ 1.3a	13.95 $\pm$ 2.3b
	D	27.17 $\pm$ 4.7a	27.79 $\pm$ 0.9a	37.88 $\pm$ 3.2b	30.08 $\pm$ 2.2b
Mg	F	2.67 $\pm$ 0.04a	3.41 $\pm$ 0.04a	4.36 $\pm$ 0.4b	4.53 $\pm$ 0.05b
	D	4.27 $\pm$ 0.07a	5.07 $\pm$ 0.07b	4.83 $\pm$ 0.5a	6.03 $\pm$ 0.07c
P	F	8.69 $\pm$ 0.9a	9.56 $\pm$ 0.03a	9.45 $\pm$ 0.7a	12.95 $\pm$ 1.2b
	D	12.13 $\pm$ 0.78a	16.12 $\pm$ 1.2b	19.89 $\pm$ 1.3c	18.98 $\pm$ 2.2c
K	F	20.70 $\pm$ 1.1a	21.33 $\pm$ 1.2a	23.67 $\pm$ 2.1a	28.18 $\pm$ 1.4b
	D	275.84 $\pm$ 10.4a	246.74 $\pm$ 14.3b	361.17 $\pm$ 9.8c	173.51 $\pm$ 7.8d
Ca	F	19.08 $\pm$ 2.1a	21.60 $\pm$ 1.3a	23.02 $\pm$ 3.1a	28.10 $\pm$ 1.4b
	D	23.90 $\pm$ 1.7a	26.79 $\pm$ 2.1a	35.60 $\pm$ 1.3b	30.98 $\pm$ 2.5c
S	F	57.75 $\pm$ 2.8a	60.97 $\pm$ 3.7a	70.39 $\pm$ 4.3b	85.21 $\pm$ 2.1b
	D	72.66 $\pm$ 7.8a	84.63 $\pm$ 4.3b	104.34 $\pm$ 8.9c	100.78 $\pm$ 3.1c
Fe	F	7.95 $\pm$ 0.01a	8.78 $\pm$ 0.9b	9.63 $\pm$ 1.2b	11.70 $\pm$ 1.2c
	D	10.65 $\pm$ 0.3a	11.95 $\pm$ 0.2a	15.45 $\pm$ 2.4b	14.68 $\pm$ 2.2b

Explanations: SD = standard deviation, F = feedstock, D = digestate; within the same line values marked with the same letter are not statistically different (ANOVA, $p < 0.05$, Tukey test).

Source: own study.

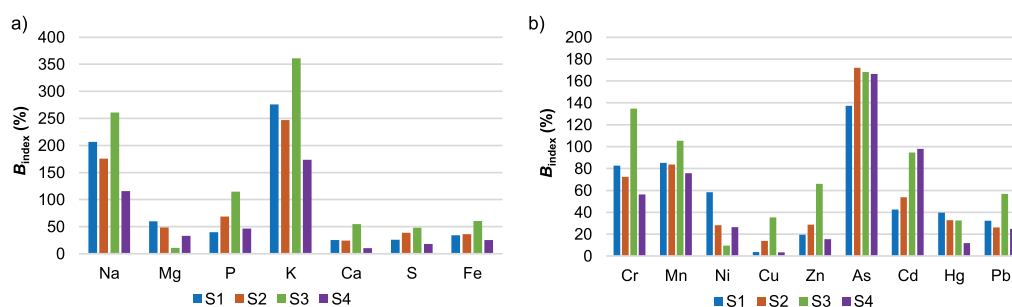


Fig. 2. The values of biodegradability index (B_{index}) with regard to macro- (a) and micro-elements (b) in experimental trials S1–S4; source: own study

Table 3. The distribution of heavy metals (HMs) in experimental trials S1–S4 ($\text{mg}\cdot\text{kg}^{-1}$ DM)

Heavy metal	Description	S1	S2	S3	S4
		mean $\pm$ SD			
Cr	F	27.57 $\pm$ 1.3a	20.21 $\pm$ 1.7b	17.83 $\pm$ 1.3c	16.89 $\pm$ 1.3c
	D	30.83 $\pm$ 1.7a	31.73 $\pm$ 1.9a	47.45 $\pm$ 2.2b	43.07 $\pm$ 2.1b
Mn	F	78.89 $\pm$ 10.7a	89.43 $\pm$ 7.8b	94.22 $\pm$ 4.5c	113.41 $\pm$ 14.2d
	D	146.04 $\pm$ 7.9a	164.13 $\pm$ 9.9b	193.44 $\pm$ 8.9c	199.39 $\pm$ 15.1c
Ni	F	25.12 $\pm$ 2.1a	20.17 $\pm$ 2.2b	20.35 $\pm$ 2.3b	18.81 $\pm$ 2.1b
	D	39.79 $\pm$ 3.3a	25.86 $\pm$ 2.4b	28.86 $\pm$ 3.3b	23.77 $\pm$ 1.4c
Cu	F	253.08 $\pm$ 2.5a	162.04 $\pm$ 14.5b	175.27 $\pm$ 10.1b	163.48 $\pm$ 1.5b
	D	262.00 $\pm$ 2.5a	184.69 $\pm$ 8.9b	237.30 $\pm$ 14.1c	168.99 $\pm$ 7.8d
Zn	F	741.20 $\pm$ 14.7a	543.36 $\pm$ 14.3b	588.27 $\pm$ 14.5c	513.48 $\pm$ 11.4d
	D	886.33 $\pm$ 12.4a	699.37 $\pm$ 12.4b	976.62 $\pm$ 16.1c	592.93 $\pm$ 17.5d
As	F	1.84 $\pm$ 0.08a	1.34 $\pm$ 0.04b	1.48 $\pm$ 0.01c	1.23 $\pm$ 0.04d
	D	4.36 $\pm$ 0.07a	3.64 $\pm$ 0.1b	3.96 $\pm$ 0.04c	3.27 $\pm$ 0.01d
Cd	F	0.85 $\pm$ 0.05a	0.72 $\pm$ 0.02b	0.65 $\pm$ 0.01c	0.60 $\pm$ 0.01d
	D	1.21 $\pm$ 0.01a	1.10 $\pm$ 0.01b	1.20 $\pm$ 0.01c	1.19 $\pm$ 0.02c
Hg	F	10.19 $\pm$ 1.1a	8.9 $\pm$ 1.3a	11.4 $\pm$ 2.1a	12.07 $\pm$ 2.1a
	D	15.24 $\pm$ 1.4a	11.83 $\pm$ 1.3b	15.1 $\pm$ 3.3a	13.49 $\pm$ 3.7a
Pb	F	11.34 $\pm$ 1.2a	8.27 $\pm$ 1.3b	9.92 $\pm$ 1.2b	7.70 $\pm$ 0.1c
	D	14.99 $\pm$ 2.1a	10.43 $\pm$ 0.9b	15.56 $\pm$ 2.2a	9.62 $\pm$ 0.3b

Explanations as in Tab. 2.

Source: own study.

drops exceeding 30% were found with regard to Cr, Cu, Cd, Zn and Pb, respectively.

Similarly as in the case of macro-nutrients, the increase of HMs content in the digestate occurred (Tab. 3). This effect was related with the weight loss resulting from organic matter decomposition as well as release of biogas within the AD process (Dong *et al.*, 2013). In almost cases, as compared to MSS mono-digestion the reduction of HMs contents in digestate was achieved. As previously, the comparable results between mono- and co-digestion series were observed for Hg. It should be noticed that the statistically significant drops were observed in S4, in three-component co-digestion. Different trend occurred also for Cr and Mn, therein in co-digestion series the enhanced contents of those HMs were found. In this case, the major growths were noticed for S3 and S4. As compared to control, for Cr the increase of 53 and 39% was found for S3 and S4, respectively. In turn, for Mn the growths by 38% and 32% were observed. It is also worth

noting that in the case of S3, with regard to Zn, Cr and Pb the highest contents among all reactors were obtained. Regarding B_{index} the major enhancement was achieved with regard to As, Mn, Cd and Cr (Fig. 2b). As compared to control, in S4 in almost all cases the lower values of this index was achieved, the exception was found only for As and Cd, respectively. However, it is worth mentioning that as compared to MSS mono-digestion, the highest increases of HMs were observed in series S3 supplied by MSS and AP.

EFFICIENCY AND STABILITY OF AD PROCESS

The effectiveness of AD of selected substrates was evaluated on the basis of volatile solids (VS) removal as well as both biogas and methane productions. As shown in Figure 3, enhanced biogas and methane yields were achieved only in S2 and S4, as compared to control (S1).

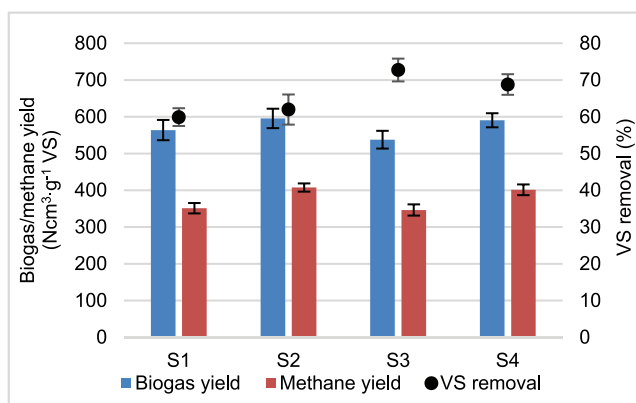


Fig. 3. Process performance in experimental trials S1–S4; VS = volatile solids; source: own study based on Szaja *et al.* (2025)

However, with regard to biogas yield, a minor increase of approx. 5% was found in both cases. A major effect was observed in the case of methane yield, therein this parameter was improved by 16 and 14% in S2 and S4, respectively. The average values of this parameter reached 407.6 and 401.6 $\text{Ncm}^3 \cdot \text{g}^{-1} \text{VS}$ in S2 and S4, respectively. However, in S3 supplied by AP, comparable results control were achieved. Therein, the methane production was established at the level of 346.4 $\text{Ncm}^3 \cdot \text{g}^{-1} \text{VS}$. This fact occurred despite the introduction of valuable macro-elements and reduction of HMs in the feedstock digester. It should be pointed out that in this case also the highest bioaccumulation of almost all HMs occurred (Fig. 2b). As it can be seen in Figure 3, in all co-digestion series the improvement of VS removal occurred compared to MSS mono-digestion. Major growths by 21.5 and 14.8% were achieved in S3 and S4, respectively. Therein, the VS removal was established at the level of 70.1 and 68% in S3 and S4, respectively.

Moreover, to complete the evaluation of process performance, stability parameters such as volatile fatty acids (VFA), alkalinity (ALK), ammonium nitrogen (AN) and pH were also analysed (Tab. 4). Moreover, the VFA to ALK ratio has been evaluated. With regard to pH value the comparable results were achieved in all experimental series, as compared to control a minor increase was achieved in the presence of ice-cream processing residues (IC) in two and three-component series. The beneficial effect in presence of IC was observed in the case of ALK; as compared to control, an improvement of this parameter occurred. A different trend was noticed in the case of S3 supplied by AP, therein this parameter was reduced by 9.8% as compared to control, it was accompanied by significantly enhanced VFA

content by 32%. This also resulted in a significant deterioration of the VFA/ALK ratio, as compared to control the increase by 44% was observed. Importantly, the AN content was established at comparable level in all experimental series.

ENVIRONMENTAL RISK

The presented HMs in digestate pose a serious environmental risk because of their potential bioaccumulation in plants and animals. Therefore, taking into account the agricultural use of MSS, its content should be controlled. Table 5 summaries the permissible levels of HMs in MSS in various countries that are planned to be introduced into the environment. Comparing the values obtained in this study with the standards in various countries, significantly lower values were obtained in all analysed cases. The exception was noticed only in the case of Zn, therein a slight exceedance of the limit value for acidic soils in China was noted. In Polish conditions the main sources of Zn are mainly household agents. Moreover, its occurrence is related with distribution of storm-water into MWWTPs. The steel structures, including roofing materials, contain Zn that leaches with rain and is discharged to combined sewers (Kujawska *et al.*, 2023).

DISCUSSION

DISTRIBUTION OF MICRO- AND MACRO-ELEMENTS WITHIN THE AD PROCESS

Importantly, the enrichment of feedstock composition in co-substrate presence is particularly beneficial in terms of effective process performance. The balanced content of macro-elements in feedstock significantly affected the growth and activity of microorganisms within the apple pomace (AD) process that might result in enhanced methane production (Kadam *et al.*, 2022). Importantly, the significant improvement of feedstock composition was found in S4 (three-component co-digestion). The analysed in this study macro-elements play a key role in the effective performance of AD process. Phosphorus is essential for energy metabolism in cells (e.g., adenosine triphosphate), as well as for the formation of nucleic acids and phospholipids. In turn, sulphur is a key element in the synthesis of certain amino acids (like cysteine and methionine) and coenzymes, which are important for microbial metabolism. Calcium also plays an important role in the AD process; it provides an adequate buffer

Table 4. The stability parameters in digestates

Experimental trial	pH	ALK	VFA	AN	VFA/ALK
		mean $\pm$ SD (mg·dm ⁻³)			
S1	7.33 $\pm$ 0.02a	4896 $\pm$ 141a	224.5 $\pm$ 14a	82.75 $\pm$ 3.3a	0.0459a
S2	7.38 $\pm$ 0.05a	4917 $\pm$ 174a	225 $\pm$ 7.9a	77.75 $\pm$ 4.1a	0.0458a
S3	7.33 $\pm$ 0.07a	4465 $\pm$ 187b	295 $\pm$ 11b	79.95 $\pm$ 6.2a	0.0661b
S4	7.38 $\pm$ 0.04a	4916 $\pm$ 172a	237 $\pm$ 14a	80.95 $\pm$ 7.8a	0.0482a

Explanations: ALK = alkalinity, VFA = volatile fatty acids, AN = ammonium nitrogen; SD = standard deviation; within the same column values marked with the same letter are not statistically different (ANOVA, $p < 0.05$, Tukey test).

Source: own study based on Szaja *et al.* (2025).

Table 5. The permissible heavy metal contents in sewage sludge for natural applications (mg·kg⁻¹ DM)

Heavy metals	This study	Poland	EU	United States	China	
					pH < 6.5	pH ≥ 6.5
Cd	1.1–1.3	20	20–40	85	5	20
Ni	23.8–39.8	300	300–400	420	100	200
Cr	20.7–47.5	500	–	1200	600	1000
Pb	9.6–15.6	750	750–1200	740	300	1000
Cu	169–262	1000	1000–1750	4300	250	500
Zn	592.0–976.6	2500	2500–4000	7500	500	1000
Hg	11.8–15.8	16	–	17	–	–
As	3.3–4.4	–	–	75	75	75
B	–	–	–	–	150	150

Source: own study based on: Council Directive (1986); EPA (1993); Ministry of Environmental Protection of the People's Republic of China (2002); Obwieszczenie (2022).

capacity in the digester. It is responsible for preventing the accumulation of volatile fatty acid (VFA). Magnesium acts as an essential co-factor for many enzymes involved in the anaerobic digestion process. The previous studies indicated that magnesium and sodium play an important role in the activity of methanogenic archaea, responsible for methane production in anaerobic digesters. Another crucial trace element is Fe; it is involved in energy metabolism. It is a co-factor of enzymes, ferredoxin and cytochrome in methanogen electron transport. It is applied in an electron acceptor and donor the conversion of CO₂ to CH₄ by the methanogenic bacteria (Myszograj, Stadnik and Pluciennik-Koropczuk, 2018; Kadam *et al.*, 2022). Previous research indicated that its low and moderate concentrations improve significantly biogas production. The values exceeding 12,000 mg·dm⁻³ are considered as inhibitory (Khatri *et al.*, 2015; Andriamanohiarisoamanana *et al.*, 2018).

With regard to digestate in co-substrate presence also the improvement of macro-elements occurred. Such upgrading of nutrients contents is particularly beneficial, taking into account further agricultural use of the digestate. The desired effect is an increase in the N, P, Ca, K and Mg, the content of which is crucial for plant growth (Nascimento *et al.*, 2020). The application of municipal sewage sludge (MSS) to soils may be profitable and constitute an efficient alternative to the commonly applied methods e.g. mineral fertilisation (it can be used as a material with fertilising properties). Such action may allow for reducing, among others, the consumption of some inorganic fertilisers for agricultural purposes. Moreover, considering that digestate is a product of the fermentation process carried out as part of the processing and disposal of MSS, it is worth pointing out the reduction of carbon dioxide emissions and lower energy inputs for obtaining material with fertilising properties (Pajura, Masłoń and Czarnota, 2023). However, the application involves many risks, mainly related to presence of undesirable substances, such heavy metals (HMs), antibiotics, pathogens, and other antimicrobial substances (Buta *et al.*, 2021; Ekane, Barquet and Rosemarin, 2021). Therefore, it is necessary to control the composition of sewage sludge discharged into the environment.

Importantly, the introduction of co-substrates to feedstock led to diminishing HMs content. The major effect was found in

case of Cr, Cu, Cd, Zn and Pb. The previous study indicated that the mentioned HMs present the most toxic effect on the AD microbes. However, the highest toxicity towards methanogens indicated by Cd; in the case of this HM, even low concentrations below 1 mg·dm⁻³ might affect methanogenic activity. Only in the case of Mn, the increased content was noted in the presence of selected additives. However, it is an essential trace element responsible for stabilisation of methyltransferase in methane-producing bacteria (Huang *et al.*, 2022; Pilarska *et al.*, 2024). Previous reports indicated that its supplementation at low concentration significantly improve biogas production (Cai *et al.*, 2018). In turn, the high levels of Cr might inhibit methanogens activity. However, As might also affect all stages in the AD process. In the case of the Acidogenesis and Acetogenesis phases, it may become less efficient, resulting in the accumulation of unprocessed organic material and intermediate by-products. It also has a negative impact on methanogenic archaea that are responsible for methane production.

EFFICIENCY AND STABILITY OF AD PROCESS

Importantly, the analysed macro-elements did not exceed the values considered to inhibit the fermentation process. Nevertheless, they may act synergistically or antagonistically with each other, in this way affecting methane generation. In S2 and S4 the major improvement of methane production was found. This fact was related with supplementation of feedstock in co-substrates rich in essential micro- and macro-elements that stimulate the methanogens (Mu *et al.*, 2019). The previous studies indicated that introduction of co-substrates rich in deficient trace elements to main substrate might improve substrate degradation and boost methane production by progressing electron transfer within AD (Qiao *et al.*, 2025). Ensuring the nutritional balance, as well as buffering capacity of digesters these are the main benefits of using substrates that translate into increased methane production (Li *et al.*, 2024). In this study, the beneficial effect was observed with regard to Mg, Ca and AN. Due to enhanced Ca and Mg contents in the presence of a co-substrate in all co-digestion series, the stability parameters were established at favourable level for methanogens (Walter *et al.*, 2022). In co-digestion series, low

ammonium nitrogen (AN) concentration and high levels of alkalinity (ALK) were achieved. The presence of both macro-elements improves the tolerance to AN. The previous studies demonstrated that in the presence of Mg decreased AN content was achieved due to struvite formation (Wang *et al.*, 2023). In turn, Ca ensures buffering capacity in digester; moreover, the latest studies indicated that this macro-element might strengthen dehydrogenases and protein-binding structure (Sun *et al.*, 2022).

However, it should be highlighted that in S3 with diminished methane production the enhanced bioaccumulation of Na, K, P and Ca was observed, the combination of those macro-elements might result in an osmotic stress that might have negatively influenced methanogens. Enhanced contents of Na, K and Ca might interfere with the uptake of other essential ions e. g. Na inhibits K transport, in turn Ca^{2+} interferes with Mg an important enzyme cofactor that might lead to reduced metabolic efficiency (Chen, Cheng and Creamer, 2008). Moreover, this fact might also lead to deterioration of VFA/ALK ratio and enhanced VFA content in S3.

In AP presence also the highest bioaccumulation of almost all HMs was found. Previous studies indicated that in particular the increased content of Hg, Pb and Cr might decrease the efficiency of the AD process resulting from methanogenic bacteria inhibition (Abdel-Shafy and Mansour, 2014). Another reason of the decreasing methane content might be related with the presence of inhibitory substances in AP, such as pectin and phenol compounds that exhibited the antioxidant and antimicrobial properties (Ampese *et al.*, 2024). Importantly, AP indicated the lignocellulosic structure that makes it challenging for enzymes and microbial agents to access cellulose.

However, the previous studies indicated that AP might be applied successfully in AD process. In the investigation conducted by Mu *et al.* (2019) the batch anaerobic co-digestion of AP and vegetable waste at different mixing TS ratios were examined. The most beneficial results was found at ratio of 2:3, therein the methane yield reached the value of $336 \text{ Ncm}^3\text{g}^{-1} \text{ VS}$. Such beneficial results were related with balanced carbon to nitrogen ratio C/N, diminished accumulation of volatile fatty acids, balanced pH value. In turn, Elsayed, Diab and Soliman (2021) investigated the co-digestion of primary sludge with various fruit and vegetable wastes including AP. In this case the most beneficial results were found at the mixing ratio of 50:50. In this study the methane yield was established at the level of $141 \text{ Ncm}^3\text{g}^{-1} \text{ VS}$, that was higher value than single-component fermentation of each substrate. Maragkaki *et al.* (2018) investigated that application of small amount of various agro-industrial by-products might led to increase methane production from 8% to even 740% in glycerol presence. Synergistic effect in co-digestion of sewage sludge and organic wastes was also noticed by Xie, Wickham and Nghiem (2017). Therein, the methane yields were 159, 652 and $157 \text{ Ncm}^3\text{g}^{-1} \text{ VS}$ in mono-digestion of primary sludge, food waste and paper pulp reject, respectively. Other studies indicated that co-digestion of MSS with food processing wastes mixed at the ratio of 50:50 might varied between wide range of $141\text{--}370 \text{ Ncm}^3\text{g}^{-1} \text{ VS}$ (Elsayed, Diab and Soliman, 2021). Demirel *et al.* (2013) indicated that co-digestion of wastewater with ice-cream production residues with mixed at the ratio of 9:1 resulted in significantly enhanced methane production as compared to control trail. Another beneficial effect observed in selected food processing wastes in improvement of organic compounds removal. Boosting of

degradation efficiency was related with the availability of organic compounds and high nutrient content that led to increased microbes activity and hence higher degradation rate (Mu *et al.*, 2020). This effect was also observed in study performed by Elsayed, Diab and Soliman (2021). Therein, in co-digestion of primary sludge with fruit and vegetable wastes the volatile solids (VS) removal varied between 63 to 73%, while the highest values were observed at the mixing ratio of 70:30 and 50:50. Importantly, the increasing the dose did not resulted in an increased methane production or improved process efficiency. Similarly, high VS removal efficiency above 60% was noticed in other studies investigated co-digestion of sewage sludge and fruit and vegetable wastes (Liu *et al.*, 2012; Maragkaki *et al.*, 2018).

ECOLOGICAL RISK

The agricultural use of MSS might offer several benefits, such as providing valuable nutrients to soil as well as reducing waste disposal costs. However, there are ecological risks related in particular with HMs presence. Among HMs, Cd, Pb and Hg indicate a major toxic effect to humans; however, their impact on plants are relatively low (Janaszek *et al.*, 2024). Their presence in the environment might cause diseases including brain damage and disfunction of essential organs e.g. lungs, kidneys, and liver (Musah, 2024). The exposure to Cd can cause various diseases, e.g. proteinuria, glycosuria, or emphysema (Kujawska *et al.*, 2024). In turn, Hg indicates the toxic effects on humans the nervous, digestive and immune systems (Ha *et al.*, 2017). The exposure to Pb might lead to central nervous system illness, peripheral neuropathy, central nervous system disorders, and anaemia (Kujawska *et al.*, 2024). In turn, the increased contents of Ni, Zn and Cu are recognised as extremely harmful to plants (Janaszek *et al.*, 2024). They inhibit photosynthesis, disturb enzyme function, affected protein synthesis, as well as interrupt nutrient transport in the plant (Moya, Ros and Picazo, 1993). Importantly, in this study the HMs contents were within the assumed standards. Another important factor that should be highlighted is fact that comparing the obtained results with other digestates from various wastewater treatment plants (WWTPs) relatively low content of HMs was achieved (Czatkowska *et al.*, 2025).

CONCLUSIONS

In this study, micro- and macro-elements variation in anaerobic co-digestion of municipal sewage sludge (MSS) and food processing wastes represented by ice-cream processing residues (IC), apple pomace (AP) orange wastes (OW) was evaluated. The obtained results indicated that the introduction of co-substrates led to a decrease in the heavy metals (HMs) content in the feedstock; simultaneously, an increase in macronutrients was achieved. The most beneficial results were observed in three-component series in the presence of MSS, IC and OW, therein enhanced methane production accompanied by improved organic compounds removal was noticed. The opposite trend, i.e. decreased methane production, was obtained in the reactor supplied by AP, therein also the highest bioaccumulation of almost all HMs occurred. However, comparing the values obtained in this study with the standards in the EU, significantly lower values were obtained, indicating the possibility of further use of all digestates.

CONFLICT OF INTEREST

All authors declare that they have no conflict of interests.

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